



HK IMO Training 2023-2024
Problem Set 4 Suggested Solutions



Problem 13 (Indonesia MO 2017 Problem 2). There are 5 points on the plane, with no three collinear. Some pairs of points are joined by a segment. We say that an ordered triple (A, B, C) of distinct points is good if one of the following holds:

- A and B are joined by a segment, and B and C are joined by a segment;
- A and B are not joined by a segment, and B and C are not joined by a segment.

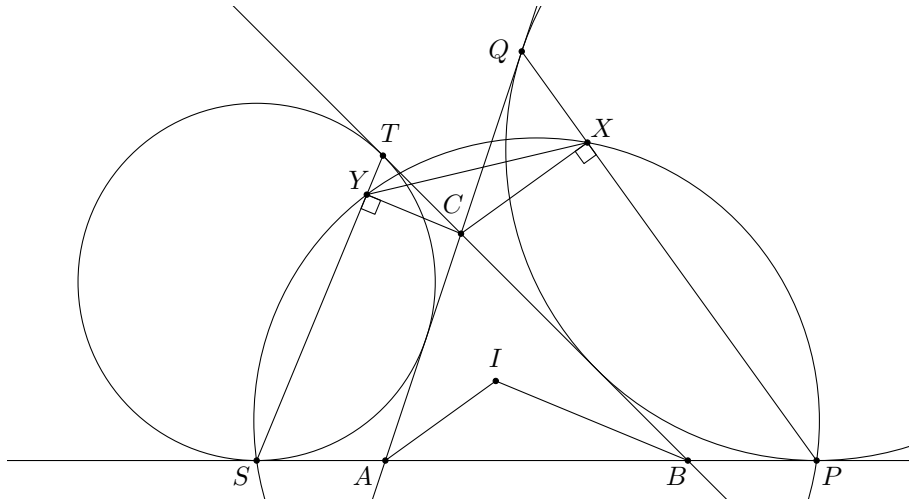
Find the minimum number of good triples.

Solution. Answer: 20.

We count the smallest number of good ordered triples formed by any 3 distinct points A, B, C . If there are 0 or 3 segments among A, B, C , then all $3! = 6$ permutations of these points form good triples. If there is 1 or 2 segments among A, B, C , say only A and B are joined by a segment, then only the triples (A, C, B) and (B, C, A) are good. Thus, every 3 distinct points form at least 2 good triples. As a result, there are at least $2 \times \binom{5}{3} = 20$ good triples in total.

For example, if only the segments AB, BC, CD, DE, EA are joined, then there is exactly 1 or 2 segments among any 3 points, and hence there are 20 good triples in total. □

Problem 14 (Balkan MO 2013 Problem 1). In $\triangle ABC$, the A -excircle touches the extensions of AB and AC at P and Q respectively. The B -excircle touches the extensions of BA and BC at S and T respectively. Let X and Y be the feet of perpendiculars from C to PQ and ST respectively. Prove that P, X, Y, S are concyclic.



Solution. Let I be the incentre of $\triangle ABC$. Note that

$$\angle XCY = \angle XCQ + \angle QCT + \angle TCY = \frac{A}{2} + C + \frac{B}{2} = 90^\circ + \frac{C}{2} = \angle BIA$$

and

$$\frac{CX}{CY} = \frac{CQ \cos \angle XCQ}{CT \cos \angle YCT} = \frac{(s-b) \cos \frac{A}{2}}{(s-a) \cos \frac{B}{2}} = \frac{s-b}{\cos \frac{B}{2}} \cdot \frac{\cos \frac{A}{2}}{s-a} = \frac{IB}{IA}.$$

This implies $\triangle CXY \sim \triangle IBA$. Therefore,

$$\angle SYX + \angle SPX = (90^\circ + \angle CYX) + \left(90^\circ - \frac{A}{2}\right) = 90^\circ + \angle IAB + 90^\circ - \frac{A}{2} = 180^\circ.$$

This proves P, X, Y, S are concyclic. □

Problem 15 (Indonesia MO 2019 Problem 8). Let $n > 1$ be a positive integer. Suppose $a_1, a_2, \dots, a_{2n}$ are integers with sum $n + 1$ such that $|a_j| \leq n$ for every j . Prove that some of $a_1, a_2, \dots, a_{2n}$ have sum 0.

Solution. We claim that there is a permutation $b_1, b_2, \dots, b_{2n}$ of $a_1, a_2, \dots, a_{2n}$ such that

$$-n + 2 \leq \sum_{i=1}^k b_i \leq n + 1 \quad (1)$$

for all $k = 1, 2, \dots, 2n$. Firstly, since $a_1 + a_2 + \dots + a_{2n} > 0$, there exists a nonnegative term. Let b_1 be any such term. Clearly, (1) holds when $k = 1$ as $0 \leq b_1 \leq n$. Suppose we have defined $b_1, b_2, \dots, b_m$ where $m < 2n$.

- If $b_1 + b_2 + \dots + b_m \leq 1$, then the sum of the remaining numbers is at least n . Thus, there exists a nonnegative term. Let b_{m+1} be any such term. Then we have

$$\begin{aligned} b_1 + b_2 + \dots + b_m + b_{m+1} &\leq 1 + n, \\ b_1 + b_2 + \dots + b_m + b_{m+1} &\geq b_1 + b_2 + \dots + b_m \geq -n + 2. \end{aligned}$$

- If $b_1 + b_2 + \dots + b_m \geq 2$ and there is a negative term among the remaining numbers, we let b_{m+1} be any such term. Then we have

$$\begin{aligned} b_1 + b_2 + \dots + b_m + b_{m+1} &< b_1 + b_2 + \dots + b_m \leq n + 1, \\ b_1 + b_2 + \dots + b_m + b_{m+1} &\geq 2 - n. \end{aligned}$$

- If $b_1 + b_2 + \dots + b_m \geq 2$ and all remaining numbers are nonnegative, we let b_{m+1} be any of the remaining terms. Then we have

$$\begin{aligned} b_1 + b_2 + \dots + b_m + b_{m+1} &= (n + 1) - \text{sum of remaining terms} \leq n + 1, \\ b_1 + b_2 + \dots + b_m + b_{m+1} &\geq b_1 + b_2 + \dots + b_m \geq 2 > -n + 2. \end{aligned}$$

In any case, the relation (1) holds when $k = m + 1$. Thus, our claim holds. Now, consider the $2n + 1$ numbers

$$0, b_1, b_1 + b_2, \dots, b_1 + b_2 + \dots + b_{2n}.$$

By (1), all of them belong to $\{-n + 2, -n + 3, \dots, n + 1\}$. There are $2n$ possibilities, and hence two of them must be the same by

the pigeonhole principle. Let $\sum_{i=1}^k b_i = \sum_{i=1}^{\ell} b_i$ for some $0 \leq k < \ell \leq 2n$. Then

$$b_{k+1} + b_{k+2} + \dots + b_{\ell} = \sum_{i=1}^{\ell} b_i - \sum_{i=1}^k b_i = 0,$$

which proves the result. □

Problem 16 (Balkan MO 2015 Problem 4). Let $\{x\} = x - \lfloor x \rfloor$ denote the fractional part of a real number x . Prove that for any 20 consecutive positive integers, we can find one of them, say n , such that

$$k\sqrt{n} \{k\sqrt{n}\} > \frac{5}{2}$$

for all positive integers k .

Solution. Among any 20 consecutive positive integers, there must be one of them, say n , which is of the form $20m + 15$. We claim that this n works.

Firstly, we prove that $k^2n - \ell^2$ cannot be 0, 1, 2, 3, 4.

- As $n \equiv 3 \pmod{4}$, n cannot be a perfect square. Thus, $k^2n - \ell^2 \neq 0$.
- If $k^2n - \ell^2 = 2, 3$, we have $-\ell^2 \equiv \pm 2 \pmod{5}$, and hence $\ell^2 \equiv \pm 2 \pmod{5}$. This is a contradiction.
- If $k^2n - \ell^2 = 1, 4$, note that n has a prime divisor p which is congruent to 3 modulo 4 since $n \equiv 3 \pmod{4}$. This implies $-\ell^2 \equiv 1, 4 \pmod{p}$, and hence $\ell^2 \equiv -1, -4 \pmod{p}$. However, it is well-known that $\left(\frac{-4}{p}\right) = \left(\frac{-1}{p}\right) = -1$. This is a contradiction.

Now, for any positive integer k , we have

$$\begin{aligned}
k\sqrt{n} \{k\sqrt{n}\} &= k\sqrt{n} (k\sqrt{n} - \lfloor k\sqrt{n} \rfloor) \\
&= k\sqrt{n} \times \frac{(k\sqrt{n})^2 - \lfloor k\sqrt{n} \rfloor^2}{k\sqrt{n} + \lfloor k\sqrt{n} \rfloor} \\
&= \frac{k\sqrt{n}}{k\sqrt{n} + \lfloor k\sqrt{n} \rfloor} \times (k^2n - \lfloor k\sqrt{n} \rfloor^2) \\
&> \frac{1}{2} (k^2n - \lfloor k\sqrt{n} \rfloor^2)
\end{aligned}$$

since $\sqrt{n}$ is irrational. But then we have shown that $k^2n - \lfloor k\sqrt{n} \rfloor^2$ cannot be 0, 1, 2, 3, 4. Therefore, we must have

$$\frac{1}{2} (k^2n - \lfloor k\sqrt{n} \rfloor^2) \geq \frac{5}{2}.$$

This completes the proof. □